

DATABASE SYSTEM ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT

Week – 2 Practical Lab ex

Experiment: BookFlow Library Management System

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Section: 5

AIM

To create a Library Management Database (BookFlow) using MySQL, implement constraints, perform JOIN and GROUP BY operations, demonstrate ACID Transactions, and improve search performance using Indexing.

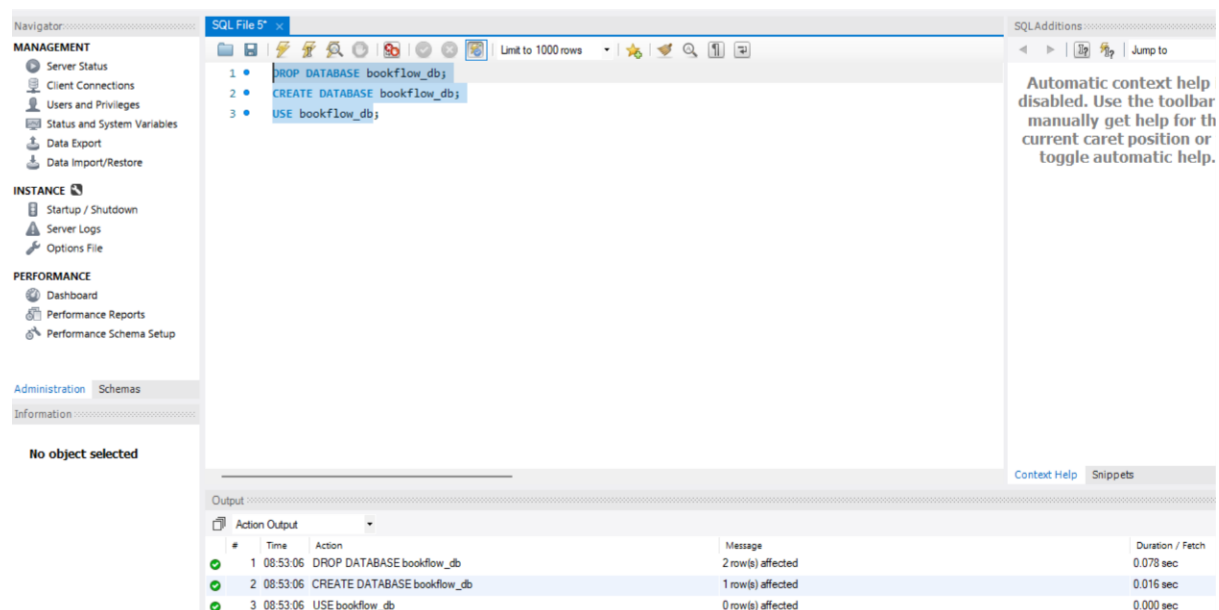
Step 1: Create the Database

SQL Query

```
CREATE DATABASE bookflow_db;
```

```
USE bookflow_db;
```

OUTPUT:



The screenshot displays the MySQL Workbench interface. The central SQL editor shows three queries: 1. DROP DATABASE bookflow_db; 2. CREATE DATABASE bookflow_db; 3. USE bookflow_db;. The bottom Output pane shows the execution results for these queries.

#	Time	Action	Message	Duration / Fetch
1	08:53:06	DROP DATABASE bookflow_db	2 row(s) affected	0.078 sec
2	08:53:06	CREATE DATABASE bookflow_db	1 row(s) affected	0.016 sec
3	08:53:06	USE bookflow_db	0 row(s) affected	0.000 sec

Step 2: Create the Books Table

SQL Query

```
CREATE TABLE Books (  
    book_id INT PRIMARY KEY,  
    title VARCHAR(100) NOT NULL,  
    isbn VARCHAR(20) UNIQUE,  
    published_year INT CHECK (published_year < 2027)  
);
```

OUTPUT:

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'Server Enterprise1' tree with 'Databases' expanded. The central pane shows a SQL script with four queries: 1. DROP DATABASE bookflow_db; 2. CREATE DATABASE bookflow_db; 3. USE bookflow_db; 4. CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(20) UNIQUE, published_year INT CHECK (published_year < 2027));. The right pane shows the 'SQL Additions' tab with a message: 'Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.' The bottom pane shows the 'Output' window with the following results:

#	Time	Action	Message	Duration / Fetch
1	08:53:06	DROP DATABASE bookflow_db	2 row(s) affected	0.078 sec
2	08:53:06	CREATE DATABASE bookflow_db	1 row(s) affected	0.016 sec
3	08:53:06	USE bookflow_db	0 row(s) affected	0.000 sec
4	08:55:11	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT...	0 row(s) affected	0.047 sec

Step 3: Insert Records into Books Table

SQL Query

```
INSERT INTO Books (book_id,title,isbn,published_year)  
VALUES  
(1,'The Great Gatsby','9780743273565',1925),  
(2,'To Kill a Mockingbird','9780061120084',1960),
```

```
(3,'1984','9780451524935',1949);
```

```
SELECT * FROM Books;
```

OUTPUT:

The screenshot displays the SQL Server Enterprise Manager interface. The left pane shows the 'Server Explorer' with the 'Books' database selected. The central pane shows the SQL query editor with the following code:

```
4 CREATE TABLE Books (  
5     book_id INT PRIMARY KEY,  
6     title VARCHAR(100) NOT NULL,  
7     isbn VARCHAR(20) UNIQUE,  
8     published_year INT CHECK (published_year < 2027)  
9 );  
10 INSERT INTO Books (book_id,title,isbn,published_year)  
11 VALUES (1,'The Great Gatsby',9780743273565,1925)  
12  
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95  
96  
97  
98  
99  
100
```

The right pane shows the 'Result Grid' with the following data:

book_id	title	isbn	published_year
1	The Great Gatsby	9780743273565	1925
2	To Kill a Mockingbird	9780061120084	1960
3	1984	9780451524935	1949

The bottom pane shows the 'Output' window with the following messages:

#	Time	Action	Message	Duration / Fetch
1	08:53:06	DROP DATABASE bookflow_db	2 row(s) affected	0.078 sec
2	08:53:06	CREATE DATABASE bookflow_db	1 row(s) affected	0.016 sec
3	08:53:06	USE bookflow_db	0 row(s) affected	0.000 sec
4	08:55:11	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NO...	0 row(s) affected	0.047 sec
5	08:57:45	INSERT INTO Books (book_id,title,isbn,published_year) VALUES (1,'The Great Gats...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
6	08:57:45	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Step 4: Create Members Table

SQL Query

```
CREATE TABLE Members (  
    member_id INT PRIMARY KEY,  
    full_name VARCHAR(100),  
    email VARCHAR(100) UNIQUE  
);
```

Automatic context help is disabled. Use the toolbar manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

#	Time	Action	Message	Duration / Fetch
3	08:53:06	USE bookflow_db	0 row(s) affected	0.000 sec
4	08:55:11	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(20) UNIQUE, published_year INT CHECK (published_year < 2027));	0 row(s) affected	0.047 sec
5	08:57:45	INSERT INTO Books (book_id,title,isbn,published_year) VALUES (1,'The Great Gatsby','9780743273565',1925), (2,'To Kill a Mockingbird','9780061120084',1960), (3,'1984','9780451524935',1949);	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
6	08:57:45	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
7	08:59:57	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100) UNIQUE);	0 row(s) affected	0.032 sec

Step 5: Insert Records into Members Table

SQL Query

INSERT INTO Members VALUES

(101,'John Smith','john.smith@email.com'),

(102,'Emma Wilson','emma.wilson@email.com'),

(103,'Michael Brown','michael.brown@email.com');

SELECT * FROM Members;

OUTPUT:

Automatic context help is disabled. Use the toolbar manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

member_id	full_name	email
101	John Smith	john.smith@email.com
102	Emma Wilson	emma.wilson@email.com
103	Michael Brown	michael.brown@email.com

Members 2 x

Read Only Context Help Snippets

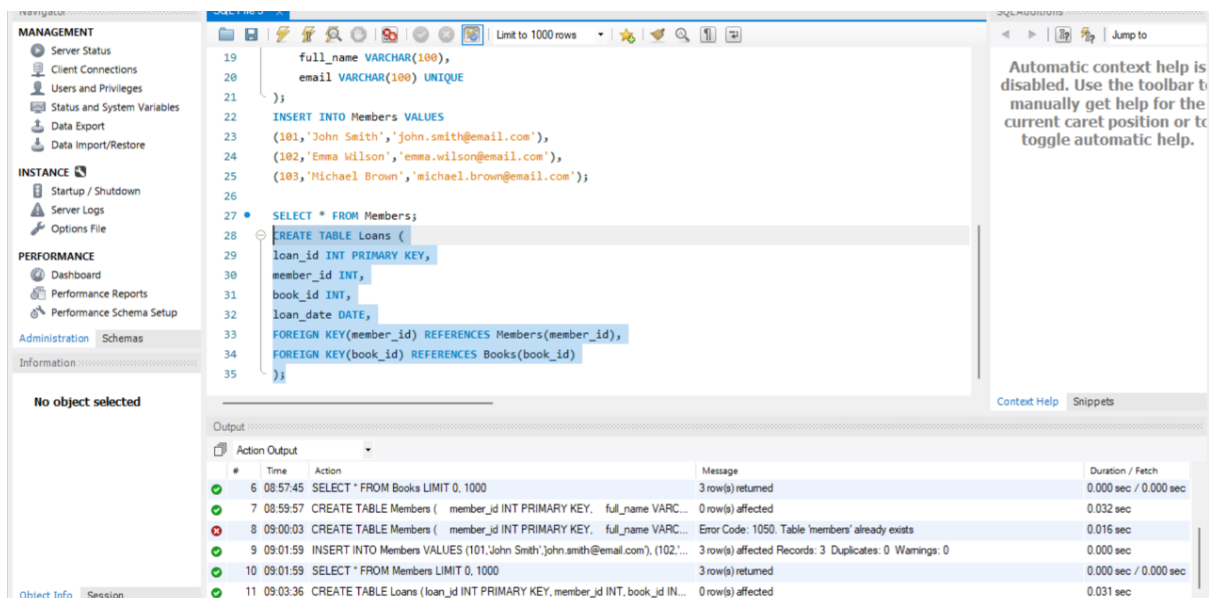
#	Time	Action	Message	Duration / Fetch
5	08:57:45	INSERT INTO Books (book_id,title,isbn,published_year) VALUES (1,'The Great Gatsby','9780743273565',1925), (2,'To Kill a Mockingbird','9780061120084',1960), (3,'1984','9780451524935',1949);	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
6	08:57:45	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
7	08:59:57	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100) UNIQUE);	0 row(s) affected	0.032 sec
8	09:00:03	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100) UNIQUE);	Error Code: 1050. Table 'members' already exists	0.016 sec
9	09:01:59	INSERT INTO Members VALUES (101,'John Smith','john.smith@email.com'), (102,'Emma Wilson','emma.wilson@email.com'), (103,'Michael Brown','michael.brown@email.com');	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
10	09:01:59	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Step 6: Create Loans Table

SQL Query

```
CREATE TABLE Loans (  
    loan_id INT PRIMARY KEY,  
    member_id INT,  
    book_id INT,  
    loan_date DATE,  
    FOREIGN KEY(member_id) REFERENCES Members(member_id),  
    FOREIGN KEY(book_id) REFERENCES Books(book_id)  
);
```

OUTPUT:



The screenshot displays the MySQL Workbench interface. The left sidebar shows the 'MANAGEMENT' and 'PERFORMANCE' tabs. The main editor window contains the following SQL queries:

```
19 full_name VARCHAR(100),  
20 email VARCHAR(100) UNIQUE  
21 );  
22 INSERT INTO Members VALUES  
23 (101,'John Smith','john.smith@email.com'),  
24 (102,'Emma Wilson','emma.wilson@email.com'),  
25 (103,'Michael Brown','michael.brown@email.com');  
26  
27 • SELECT * FROM Members;  
28 CREATE TABLE Loans (  
29 loan_id INT PRIMARY KEY,  
30 member_id INT,  
31 book_id INT,  
32 loan_date DATE,  
33 FOREIGN KEY(member_id) REFERENCES Members(member_id),  
34 FOREIGN KEY(book_id) REFERENCES Books(book_id)  
35 );
```

The 'Output' tab at the bottom shows the execution results of the queries:

#	Time	Action	Message	Duration / Fetch
6	08:57:45	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
7	08:59:57	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARC...	0 row(s) affected	0.032 sec
8	09:00:03	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARC...	Error Code: 1050. Table 'members' already exists	0.016 sec
9	09:01:59	INSERT INTO Members VALUES (101,'John Smith','john.smith@email.com'), (102...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
10	09:01:59	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
11	09:03:36	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id IN...	0 row(s) affected	0.031 sec

Step 7: Insert Records into Loans Table

SQL Query

INSERT INTO Loans VALUES

```
(1,101,1,'2025-01-05'),  
(2,102,2,'2025-01-08'),  
(3,103,3,'2025-01-10'),  
(4,101,2,'2025-02-01'),  
(5,102,1,'2025-02-05'),  
(6,103,2,'2025-02-12'),  
(7,101,3,'2025-03-01'),  
(8,102,3,'2025-03-07'),  
(9,103,1,'2025-03-15'),  
(10,101,1,'2025-04-01');
```

SELECT * FROM Loans;

OUTPUT:

The screenshot displays the SQL Server Enterprise Manager interface. The left-hand pane shows the 'Navigation' tree with 'Administration' and 'Schemas' expanded. The main window shows the 'SQL File 6*' editor with the following SQL query:

```
36 INSERT INTO Loans VALUES  
37 (1,101,1,'2025-01-05'),  
38 (2,102,2,'2025-01-08'),  
39 (3,103,3,'2025-01-10'),  
40 (4,101,2,'2025-02-01'),  
41 (5,102,1,'2025-02-05'),  
42 (6,103,2,'2025-02-12'),  
43 (7,101,3,'2025-03-01'),  
44 (8,102,3,'2025-03-07'),  
45 (9,103,1,'2025-03-15'),  
46 (10,101,1,'2025-04-01');
```

Below the query editor, the 'Result Grid' shows the output of the query, which is a table with 10 rows and 4 columns: loan_id, member_id, book_id, and loan_date.

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	102	3	2025-03-07
9	103	1	2025-03-15
10	101	1	2025-04-01

At the bottom, the 'Output' pane shows the 'Action Output' for the query, indicating that 10 rows were affected, with 0 duplicates and 0 warnings.

12 09:07:18 INSERT INTO Loans VALUES (1,101,1,'2025-01-05'), (2,102,2,'2025-01-08'), (3,103,3,'2025-01-10'), (4,101,2,'2025-02-01'), (5,102,1,'2025-02-05'), (6,103,2,'2025-02-12'), (7,101,3,'2025-03-01'), (8,102,3,'2025-03-07'), (9,103,1,'2025-03-15'), (10,101,1,'2025-04-01'); 10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0 Duration / Fetch: 0.000 sec

Step 8: INNER JOIN Query

SQL Query

```
SELECT m.full_name AS Member_Name,  
b.title AS Book_Title  
FROM Loans l  
INNER JOIN Members m  
ON l.member_id=m.member_id  
INNER JOIN Books b  
ON l.book_id=b.book_id;
```

OUTPUT:

The screenshot displays a database management interface with a sidebar on the left containing sections for MANAGEMENT, INSTANCE, PERFORMANCE, and Information. The main window shows a SQL query editor with the following code:

```
49 SELECT m.full_name AS Member_Name,  
50 b.title AS Book_Title  
51 FROM Loans l  
52 INNER JOIN Members m  
53 ON l.member_id=m.member_id
```

Below the query editor, the 'Result Grid' shows the output of the query:

Member_Name	Book_Title
John Smith	The Great Gatsby
John Smith	To Kill a Mockingbird
John Smith	1984
John Smith	The Great Gatsby
Emma Wilson	To Kill a Mockingbird
Emma Wilson	The Great Gatsby
Emma Wilson	1984
Michael Brown	1984
Michael Brown	To Kill a Mockingbird
Michael Brown	The Great Gatsby

At the bottom, the 'Output' pane shows a log of database actions:

#	Time	Action	Message	Duration / Fetch
9	09:01:59	INSERT INTO Members VALUES (101,'John Smith','john.smith@email.com'), (102,'...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
10	09:01:59	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
11	09:03:36	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id IN...	0 row(s) affected	0.031 sec
12	09:07:18	INSERT INTO Loans VALUES (1,101,1,2025-01-05), (2,102,2,2025-01-08), (3,10...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
13	09:07:18	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
14	09:10:10	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNE...	10 row(s) returned	0.016 sec / 0.000 sec

Step 9: GROUP BY Query

SQL Query

```
SELECT published_year,  
COUNT(book_id) AS Total_Books  
FROM Books
```

GROUP BY published_year

ORDER BY published_year;

OUTPUT:

The screenshot shows the SQL Enterprise Manager interface. The left pane contains the 'Navigation' tree with categories like MANAGEMENT, INSTANCE, PERFORMANCE, and Administration. The main pane displays a SQL query in a text editor, which is a JOIN query between 'Members' and 'Books' tables, grouped by 'published_year' and ordered by 'published_year'. Below the query editor, the 'Result Grid' shows the output of the query, which is a table with two columns: 'published_year' and 'Total_Books'. The results are as follows:

published_year	Total_Books
1925	1
1949	1
1960	1

Below the result grid, the 'Output' pane shows the 'Action Output' for the query, including the time taken to execute the query and the number of rows returned.

Step 10: Create Donation_History Table

SQL Query

```
CREATE TABLE Donation_History (  
    donation_id INT PRIMARY KEY,  
    book_id INT,  
    donor_name VARCHAR(100),  
    donation_date DATE,  
    FOREIGN KEY(book_id) REFERENCES Books(book_id)  
);
```

OUTPUT:

The screenshot displays the SQL Server Enterprise Manager interface. The left pane shows the 'Server Enterprise' tree with 'Databases' expanded. The center pane shows a SQL query file with the following code:

```
51 FROM Loans l
52 INNER JOIN Members m
53 ON l.member_id=m.member_id
54 INNER JOIN Books b
55 ON l.book_id=b.book_id;
56 SELECT published_year,
57 COUNT(book_id) AS Total_Books
58 FROM Books
59 GROUP BY published_year
60 ORDER BY published_year;
61 CREATE TABLE Donation_History (
62 donation_id INT PRIMARY KEY,
63 book_id INT,
64 donor_name VARCHAR(100),
65 donation_date DATE,
66 FOREIGN KEY(book_id) REFERENCES Books(book_id)
67 );
```

The right pane shows the 'SQLAdditions' tab with a message: 'Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.'

The bottom pane shows the 'Output' window with the following data:

#	Time	Action	Message	Duration / Fetch
11	09:03:36	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT)	0 row(s) affected	0.031 sec
12	09:07:18	INSERT INTO Loans VALUES (1,101,1,2025-01-05), (2,102,2,2025-01-08), (3,103,3,2025-01-11), (4,104,4,2025-01-14), (5,105,5,2025-01-17), (6,106,6,2025-01-20), (7,107,7,2025-01-23), (8,108,8,2025-01-26), (9,109,9,2025-01-29), (10,110,10,2025-02-01)	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
13	09:07:18	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
14	09:10:10	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m ON l.member_id=m.member_id INNER JOIN Books b ON l.book_id=b.book_id	10 row(s) returned	0.016 sec / 0.000 sec
15	09:12:17	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year	3 row(s) returned	0.000 sec / 0.000 sec
16	09:14:45	CREATE TABLE Donation_History (donation_id INT PRIMARY KEY, book_id INT, donor_name VARCHAR(100), donation_date DATE, FOREIGN KEY(book_id) REFERENCES Books(book_id))	0 row(s) affected	0.031 sec

Step 11: Transaction

SQL Query

START TRANSACTION;

INSERT INTO Books

VALUES

(4,'Animal Farm','9780451526342',1945);

INSERT INTO Donation_History

VALUES

(1,4,'Raj Kumar',CURDATE());

COMMIT;

SELECT * FROM Books;

SELECT * FROM Donation_History;

OUTPUT:

The screenshot displays the SQL Enterprise Manager interface. The central pane shows a SQL script with the following lines:

```
73
74 • INSERT INTO Donation_History
75 VALUES
76 (1,4,'Raj Kumar',CURDATE());
77
78 • COMMIT;
79
80 • SELECT * FROM Books;
81 • SELECT * FROM Donation_History;
```

Below the script, the 'Result Grid' shows a single row of data:

donation_id	book_id	donor_name	donation_date
1	4	Raj Kumar	2026-08-01

The 'Output' pane at the bottom shows the execution log:

#	Time	Action	Message	Duration / Fetch
17	09:16:10	START TRANSACTION	0 row(s) affected	0.000 sec
18	09:16:10	INSERT INTO Books VALUES (4,'Animal Farm','9780451526342',1945)	1 row(s) affected	0.000 sec
19	09:16:10	INSERT INTO Donation_History VALUES (1,4,'Raj Kumar',CURDATE())	1 row(s) affected	0.000 sec
20	09:16:10	COMMIT	0 row(s) affected	0.000 sec
21	09:16:10	SELECT * FROM Books LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
22	09:16:10	SELECT * FROM Donation_History LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

Step 12: Create Index

SQL Query

CREATE INDEX idx_books_isbn
ON Books(isbn);

SELECT * FROM Books

WHERE isbn='9780451524935';

OUTPUT:

MANAGEMENT

Server Status

Client Connections

Users and Privileges

Status and System Variables

Data Export

Data Import/Restore

INSTANCE

Startup / Shutdown

Server Logs

Options File

PERFORMANCE

Dashboard

Performance Reports

Performance Schema Setup

Administration

Schemas

Information

No object selected

Object Info

Session

SQL Editor

Limit to 1000 rows

SQL Editor

78 • COMMIT;
79
80 • SELECT * FROM Books;
81 • SELECT * FROM Donation_History
82 • CREATE INDEX idx_books_isbn
83 ON Books(isbn);
84
85 • SELECT * FROM Books
86 WHERE isbn='9780451524935';

Result Grid

book_id title isbn published_year

3 1984 9780451524935 1949

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Books 8 x

Read Only

Context Help

Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
19	09:16:10	INSERT INTO Donation_History VALUES (1,4,'Raj Kumar',CURDATE())	1 row(s) affected	0.000 sec
20	09:16:10	COMMIT	0 row(s) affected	0.000 sec
21	09:16:10	SELECT * FROM Books LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
22	09:16:10	SELECT * FROM Donation_History LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
23	09:17:32	CREATE INDEX idx_books_isbn ON Books(isbn)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.078 sec
24	09:17:33	SELECT * FROM Books WHERE isbn='9780451524935' LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec